

UNIVERSITY OF ZAGREB
FACULTY OF MECHANICAL ENGINEERING AND NAVAL ARCHITECTURE

BACHELOR'S THESIS

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I hereby declare that I have written this thesis independently, using the knowledge acquired during my studies and the literature cited in the bibliography.

I would like to sincerely thank my mentor, prof. dr. sc. Andrej Jokić for constructive advice and discussions, as well as for his patience while I was writing this thesis.

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ZAVRŠNI ZADATAK

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Naslov rada na
hrvatskom jeziku: **Willemsova fundamentalna lema s primjenama**Naslov rada na
engleskom jeziku: **Willems' Fundamental Lemma with Applications**

Opis zadatka:

Classical approach to simulation and controller design for dynamical systems is a model-based approach. A model of system's dynamics is obtained either directly by applying laws of physics on the considered system or indirectly via system identification. The latter is a data-based approach and the data, which consists of recorded input-output trajectories of the system, is used in a suitably defined optimization problem to obtain a model.

Alternative to model-based approaches is the direct data-based approach where simulations and/or controller synthesis is performed directly from data, completely avoiding a model. Such approach has gained significant attention in the control systems community over the past years. There are several reasons for that and the obvious one is that modern IT technologies enable collecting (sensing/recording) of large data sets in an inexpensive way. From a scientific point of view, the direct data-based approach is attractive as it can offer exact results with robustness guarantees, similar to the classical model-based approach.

One of the key results that has inspired and enabled a set of methods in direct data-based approach is the so-called Willems' fundamental lemma. Loosely speaking, this lemma states that all possible trajectories of a linear time invariant (LTI) dynamical system belong to an image space of a suitable defined data-based matrix. The main goal of this thesis is to summarize dominant state-of-the-art approaches to data-based simulation and controller design and to suitably illustrate them on several examples.

The thesis should contain the following:

- Presentation of the Willems' fundamental lemma together with discussion of its origins and meaning.
- Literature overview of the state-of-the-art results on data-based approaches which are based on the Fundamental lemma.
- On a set of suitably selected and designed examples illustrate some of the data-based approaches for simulation and control of LTI dynamical systems.
- Using suitable simulation tools (e.g., MATLAB/Simulink) to investigate applicability of the Fundamental lemma on nonlinear systems.

The thesis must contain the list of used literature as well as any assistance received.

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LIST OF SYMBOLS

Symbol	Description	Unit
$\mathbb{N}$	Set of natural numbers	
$\mathbb{R}$	Set of real numbers	
$\mathbb{Z}_{\geq 0}$	Set of non-negative integers	
$\mathbb{Z}_{> 0}$	Set of positive integers	
A, B, C, D	State-space system matrices	
A_d, B_d, C_d, D_d	Discrete-time state-space system matrices	
I	Identity matrix	
$\mathcal{C}$	Controllability matrix	
$\mathcal{O}$	Observability matrix	
Ω_L, Ω_k	Finite-horizon observability matrix	
Θ_L	Block Toeplitz matrix of Markov parameters	
M	Input–output trajectory mapping matrix in the proof	
$M_{r \times c}(\mathbb{R})$	Space of real matrices with r rows and c columns	
$(\cdot)^T$	Matrix or vector transpose	
$(\cdot)^*$	Optimal value or optimal sequence	
$\text{rank}(\cdot)$	Matrix rank	
$\ker(\cdot)$	Kernel (null space) of a matrix or linear map	
$\text{im}(\cdot)$	Image (column space) of a matrix or linear map	
$\dim(\cdot)$	Dimension of a vector space	
$x(t), x_k, x_j$	State vector at continuous time t , sample k , or prediction stage j	
$\dot{x}$	Time derivative of the state vector	
x_0	Initial state	
x_{ini}	Initial state of a trajectory	
x_s	Steady-state state vector	
$\hat{x}$	Estimated state vector	
x_f	Desired final state	
x_1, x_2	Positions of the first and second masses	m
$\dot{x}_1, \dot{x}_2$	Velocities of the first and second masses	m/s
$u(t), u_k, u_j$	Input at time t , sample k , or prediction stage j	
$\bar{u}, \bar{u}_j$	Stacked input trajectory and its value at prediction stage j	
$\bar{u}_{ini}$	Initial segment of the input trajectory	
u_{ini}, y_{ini}	Initial input and output trajectories used by DeePC	
u_s	Steady-state input	N

Symbol	Description	Unit
u^*	Optimal input sequence	
$y(t), y_k, y_j$	Output at time t , sample k , or prediction stage j	
$\bar{y}, \bar{y}_j$	Stacked output trajectory and its value at prediction stage j	
$\bar{y}_{ini}$	Initial segment of the output trajectory	
$\bar{y}^{\text{data}}$	Output trajectory obtained from data	
$\bar{y}^{\text{model}}$	Output trajectory predicted by the system model	
$\bar{y}^{\text{lin. model}}$	Output trajectory predicted by the linearized model	
y_s	Steady-state output	m
v_k	Measurement-noise sample	m
σ	Measurement-noise standard deviation	m
$\mathcal{N}(0, \sigma^2)$	Zero-mean normal distribution of the measurement noise	
σ_y	Output-trajectory slack variable	
t	Time	s
t_0, t_{ini}	Initial time or initial-trajectory time	s
T_s	Sampling period	s
τ	Time shift; applied pendulum torque	
T	Length of the recorded trajectory	
L	Trajectory length or Hankel-matrix depth	
L_{ini}, L_{ref}	Initial and reference trajectory lengths	
ℓ	Lag of the system	
n	State-space dimension	
N	State-dimension bound or prediction horizon	
m	Number of inputs	
m (pendulum)	Pendulum mass	kg
m_1, m_2	Masses in the spring–mass–damper system	kg
p	Number of outputs	
r, r_{t+j}	Reference trajectory and its value at prediction stage $t + j$	
H_L, H_k	Hankel matrix of depth L or k	
U_p, U_f	Past and future input Hankel-matrix blocks	
Y_p, Y_f	Past and future output Hankel-matrix blocks	
$X_{[0, T-L]}$	Matrix of recorded state samples	
J_L	Stacked state and input data matrix in the proof	
g (trajectory)	Coefficient vector representing a trajectory	
g (pendulum)	Gravitational acceleration	m/s ²
a	Left-kernel vector component in the proof	
b_i	Input-vector coefficient in the proof	
v	Non-zero vector in the left kernel of M	

Symbol	Description	Unit
α_j	Coefficients of the characteristic polynomial of A	
β_l	Input coefficients derived in the proof	
λ	Indeterminate of the characteristic polynomial	
$p(\lambda)$	Characteristic polynomial of A	
P	Generic positive-semidefinite weighting matrix	
Q, R	Output and input cost matrices	
$\ \cdot\ $	Norm of a vector	
$\ \cdot\ _2$	Euclidean (2-)norm of a vector	
$\ \cdot\ _2^2$	Squared Euclidean norm	
$\ \cdot\ _P^2$	Squared weighted norm induced by P , $\ z\ _P^2 = z^T P z$	
$\succeq 0$	Positive semidefinite matrix relation	
$\succ 0$	Positive definite matrix relation	
$s_{[i,j]}$	Stacked segment of sequence s from index i through j	
$\sin(\cdot)$	Sine function	
e^{At}	Matrix exponential	
$\min, \max$	Minimum and maximum operators	
$\text{col}(\cdot)$	Column-stacking operator	
$:=$	Equality by definition	
$\sim$	Distributed as, or of approximately the indicated order	
$\iff$	Logical equivalence (“if and only if”)	
$\implies$	Logical implication	
$\in$	Set-membership relation	
$\subseteq, \subsetneq$	Subset and strict-subset relations	
$\forall$	Universal quantifier (“for all”)	
$\mathcal{U}, \mathcal{Y}$	Input and output constraint sets	
λ_g, λ_y	Regularization weights	
b	Pendulum damping coefficient	N m s
c_1, c_2	Viscous damping coefficients	N s/m
k_1, k_2 (springs)	Linear spring stiffness coefficients	N/m
k_{23}	Nonlinear spring stiffness coefficient	N/m ³
k_i (RK4)	Runge–Kutta stage derivative, $i = 1, 2, 3, 4$	
E	Nonlinear spring coefficient vector	
F_2	Force exerted by the second spring	N
δl	Relative displacement of the nonlinear spring	m
l	Pendulum length; summation index in the proof	
θ	Pendulum angle	rad
$f(x, u)$	Nonlinear state-dynamics function	

LIST OF ABBREVIATIONS

Abbreviation	Description
C-H	Cayley–Hamilton theorem
COD	Complete orthogonal decomposition
DeePC	Data-enabled predictive control
LTI	Linear time-invariant
MIMO	Multiple-input, multiple-output
MPC	Model predictive control
PBH	Popov–Belevitch–Hautus test
PRBS	Pseudo-random binary sequence
QP	Quadratic program
SISO	Single-input, single-output
SVD	Singular value decomposition
ZOH	Zero-order hold

SAŽETAK

U ovom je radu obrađen rezultat poznat kao Willemsova fundamentalna lema, koji predstavlja temeljni rezultat za analizu linearnih sustava temeljenu na podacima. Najprije su obrađene teorijske osnove potrebne za razumijevanje same leme te je iznesen njezin dokaz. Potom je lema validirana na različitim numeričkim primjerima provedenima u simulacijskom okruženju. Demonstrirana je mogućnost simulacije budućeg ponašanja linearnih sustava isključivo na temelju snimljenih podataka, bez poznavanja njihova matematičkog modela. Također je istražena primjenjivost leme na određene nelinearne sustave. Lema je zatim primijenjena za sintezu prediktivnog regulatora za linearne sustave zasnovanog na podacima korištenjem optimizacijskih metoda. Regulator je dodatno nadograđen postupcima robustifikacije kako bi bio prikladniji za primjenu na stvarnim sustavima koji su pod utjecajem šuma i nelinearnosti.

KLJUČNE RIJEČI: Willemsova fundamentalna lema, analiza temeljena na podacima, simulacija budućeg ponašanja, upravljanje temeljeno na podacima, optimizacija.

ABSTRACT

This thesis provides an examination of Willems' Fundamental Lemma, a fundamental result in the data-based analysis of linear systems. First, the theoretical foundations required to understand the lemma are presented, together with its proof. The lemma is then validated through various numerical examples conducted in a simulation environment. The possibility of simulating the future behaviour of linear systems exclusively from recorded data, without any knowledge of their mathematical models, is demonstrated. The applicability of the lemma to certain nonlinear systems is also investigated. The lemma is subsequently used to design a data-based predictive controller for linear systems by applying optimization methods. The controller is further enhanced through robustification techniques to make it more suitable for practical systems affected by noise and nonlinearities.

Keywords: Willems' Fundamental Lemma, data-based analysis, simulating the future behaviour, data-based control, optimization

1 Introduction

This chapter introduces the motivation for exploring the Willems' Fundamental Lemma, which is a famous result in the data-based approach to control. Literature overview is also presented, in which we list papers and books that led to the development of the Lemma as well as some extensions and applications that are worth mentioning and are used in the rest of the thesis.

1.1 Motivation

Classical control theory, generally, assumes that a mathematical model of a system is known for controller design. That model is typically found using system identification and/or using first principles, i.e., the laws of physics. For complex systems, both of those approaches can fail to create a high-fidelity model, thus limiting the ability to control those systems with precision. An alternative to the model-based approach is the direct data-based approach to systems theory, which only uses measured data to simulate or control the system, completely avoiding the model. That approach has gained popularity since the rapid development of computers made data collection and processing relatively inexpensive. The data-based approach is a large field spanning multiple scientific disciplines, most famous is the development of machine learning, i.e., artificial intelligence. In this thesis we present a valuable result from linear systems theory which has enabled a vast number of significant results in the data-based approach. That result is called Willems' Fundamental Lemma, which was first introduced by Willems et al. [1]. Informally, it states that if a linear system is excited with a diverse enough input, then, using the recorded input-output data, it is possible to describe (parametrize) every possible combination of inputs and outputs. That is essentially the same role of a model but using only the data. To illustrate how large of an impact the Fundamental Lemma had in the last few years, since the data-based control has entered in the focus of control systems' society around the year 2020, we graphed the number of citations since its publication year until year 2025 in Figure 1.1. In the following sections we will state the famous results and explore its applications to various dynamical systems.

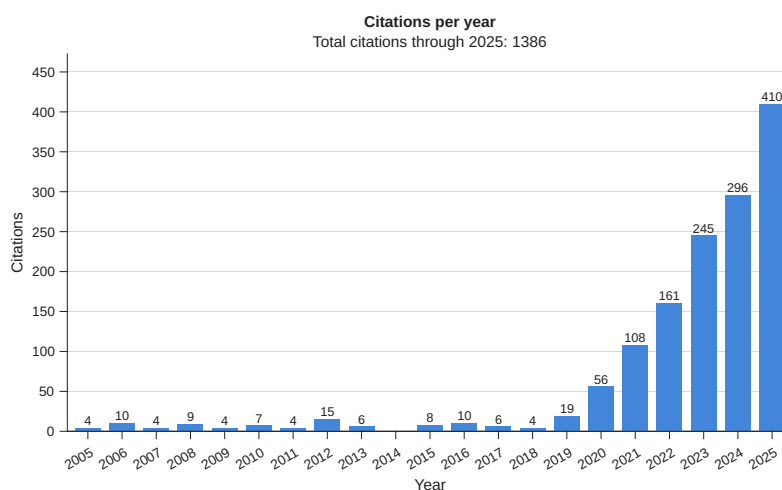


Figure 1.1: Citations per annum of [1] - Google Scholar

1.2 Literature Overview

The Fundamental Lemma is a result stemming from Jan C. Willems'¹ behavioral systems theory [2] which represents a system only with a space of its input and output trajectories. That approach was, firstly, unpopular because of its abstractness, but after a few years after the publication of the Lemma [1] it became popular. A lot of that popularity can be attributed to [3], which used the Lemma and results from the behavioural approaches to synthesise a predictive controller based entirely on data which could rival controllers from model-based approach. After that result, the applications of data-based control using behavioural approach have seen a large increase (1.1), which was nicely summarized in [4] and [5]. The book [6] gives a good overview of the main applications of the Lemma. We are largely using the formulations stated in it. The paper [7] improves the data-based controller to better suit realistic and nonlinear systems. This is the current state of the art when it comes to data-based control. Current research is highly active, so we can expect more results in the future, but the Fundamental Lemma still stands as the one result that enabled that breakthrough.

¹Jan C. Willems (1939-2013) was a Belgian systems and control theorist, best known for developing the behavioral approach to dynamical systems and for foundational contributions to dissipativity theory.

2 The Fundamental Lemma

2.1 Theoretical Background

The goal of this section is to provide an introduction to mathematical concepts needed for understanding the rest of the thesis. We will also present some standard results from linear control systems theory.

2.1.1 LTI Dynamical Systems

Although Willems' Fundamental Lemma stems from the behavioural approach to systems theory, in this thesis it is formulated using the classical state-space LTI system formulation. This formulation is simpler, yet sufficient for the remaining part of the thesis.

Definition 1. A discrete-time linear time-invariant (LTI) system is a dynamical system which admits a representation in the following state-space form:

$$x(t+1) = Ax(t) + Bu(t), \quad (2.1a)$$

$$y(t) = Cx(t) + Du(t), \quad (2.1b)$$

where $t \in \mathbb{Z}_{\geq 0}$ is the time, $x(t) \in \mathbb{R}^n$ is the state, $u(t) \in \mathbb{R}^m$ is the input and $y(t) \in \mathbb{R}^p$ is the output of the system.

The true state-space dimension is n and the system matrices $A \in M_n(\mathbb{R})$, $B \in M_{n \times m}(\mathbb{R})$, $C \in M_{p \times n}(\mathbb{R})$, $D \in M_{p \times m}(\mathbb{R})$.

In the data-based approach, which is considered in this thesis, the matrices defining the state-space model (i.e., the matrices A, B, C, D) are generally unknown. However, we will often assume that an upper bound on state-space dimension is given, i.e., we know some $N \in \mathbb{N}$ such that $N \geq n$. The concept of a system trajectory is defined next.

Definition 2. A sequence of ordered triples $(u(t), x(t), y(t))_{t=0}^{\infty}$ is called an input-state-output trajectory if

$$\begin{bmatrix} x(t+1) \\ y(t) \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} x(t) \\ u(t) \end{bmatrix}, \quad \forall t \in \mathbb{Z}_{\geq 0}.$$

It can be deduced that, from linearity of (2.1), any linear combination of input-state-output trajectories is itself an input-state-output trajectory, i.e., the set of all such trajectories spans a vector space over $\mathbb{R}$. That space is called the *behaviour* [2] of the system. Additionally, from the time invariance (2.1) it can be deduced that if $(u(t), x(t), y(t))_{t=0}^{\infty}$ is a trajectory, then:

$$(u(t+\tau), x(t+\tau), y(t+\tau))_{t=0}^{\infty}$$

is also an input-state-output trajectory for all $\tau \in \mathbb{N}$.

Definition 3. A sequence of ordered pairs $(u(t), y(t))_{t=0}^{\infty}$ is called an input-output trajectory if there exists a sequence of states $x : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}^n$ such that $(u(t), x(t), y(t))_{t=0}^{\infty}$ is an input-state-output trajectory.

In practice we are limited to working with finite data sequences, so we need to take a snippet of the trajectory relevant to us. Motivated by that we state the next definition.

Definition 4. Let $i, j \in \mathbb{Z}_{\geq 0}$ such that $i \leq j$. Given an input-state-output trajectory $(u(t), x(t), y(t))_{t=0}^{\infty}$, the sequence $(u(t), x(t), y(t))_{t=i}^j$ is called a restricted input-state-output trajectory. Restricted input-output trajectories are defined in the same way.

2.1.2 Controllability and Observability

In this section the concepts of controllability and observability are defined. Continuing in line with the state-space representation (2.1) from the previous section, we define controllability and observability in the same framework. However, although the classical and behavioral notions of controllability and observability are closely related, they are not unconditionally equivalent. Their equivalence depends on how the behavior is represented. In particular, when considering input-output behavior, equivalence with the classical state-space definitions typically requires a minimal realization; for the full input-state-output behavior, the correspondence is more direct. We will omit those comparisons between approaches, while more on that topic can be found in [8].

Definition 5. A discrete LTI system (2.1) is called controllable if for any initial state $x(0)$ and any desired state x_f , there exists a finite-length sequence of inputs

$$u(0), u(1), \dots, u(T-1)$$

such that the system reaches $x(T) = x_f$. Otherwise, the system is said to be uncontrollable.

In layman's terms, the definition above states that any state of the system can be reached from any other initial state in a finite time T by applying finitely many inputs.

Next state the famous result for testing system controllability stated and proven by Rudolf E. Kalman [9]. Let the pair (A, B) be the pair of matrices from (2.1) and let n denote the true state-space dimension of the system.

Theorem 1. (Kalman controllability theorem) The pair (A, B) is controllable if and only if the controllability matrix

$$C = \begin{bmatrix} B & AB & A^2B & \dots & A^{n-1}B \end{bmatrix}$$

is of full column rank.

The concept of observability is defined next.

Definition 6. A discrete LTI system (2.1) is called observable at $t = 0$ if there exists a finite time T so that an arbitrary initial state $x(0)$ can be uniquely reconstructed from a known restricted input-output trajectory $(u(t), x(t), y(t))_{t=0}^{T-1}$. Otherwise, the system is said to be unobservable.

Using time-invariance of (2.1), it can be shown that if the system is observable at $t = 0$, then for every $t \in \mathbb{Z}_{\geq 0}$ the state $x(t)$ is reconstructible. This result is important to control theory when it comes to state estimation (for example, Kalman filter) because it gives a necessary condition for exact state estimation.

The following theorem presents a constructive test for observability.

Theorem 2. (Kalman observability theorem) The pair (C, A) is observable if and only if the observability matrix

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

is of full row rank.

The Kalman criteria for controllability/observability were stated, but there are several other equivalent formulations. The most widely used are the Popov-Belevitch-Hautus (PBH) tests [10].

From Theorems 1 and 2, a duality between observability and controllability matrices can be seen and it is not hard to show that using matrix operations we have:

$$(A, B) \text{ is controllable} \iff (B^T, A^T) \text{ is observable.}$$

Similarly,

$$(C, A) \text{ is observable} \iff (A^T, C^T) \text{ is controllable.}$$

2.1.3 Persistency of Excitation

For the Willems' Fundamental Lemma to hold, we need the so-called persistently exciting input. Loosely speaking, a persistently exciting input is a rich and diverse enough input such that all modes of the system are expressed in the output.

Hankel Matrices

For a sequence of vectors $(s(t)_{t=i}^{t=j})$ from time $t = i$ to $t = j$, where $i < j$, we use the following abbreviation:

$$s_{[i,j]} := \begin{bmatrix} s(i) \\ s(i+1) \\ \vdots \\ s(j) \end{bmatrix}.$$

In this remainder, this notation will be used to appropriately vectorize trajectories of inputs (u), states (x) and outputs (y).

Hankel matrix is a matrix whose entries are constant on the anti-diagonals, what is contrast to Toeplitz matrix whose entries are constant on the diagonals. Informally, Hankel matrix is a mirror image of a Toeplitz matrix.

Definition 7. Given an $T \in \mathbb{N}$, let's consider a restricted input-output trajectory $(u_{[0,T-1]}, y_{[0,T-1]})$ of (2.1). Hankel matrices of depth $L \in \{1, \dots, T\}$ of these inputs and outputs are given by:

$$H_L(u_{[0,T-1]}) = \begin{bmatrix} u(0) & u(1) & \cdots & u(T-L) \\ u(1) & u(2) & \cdots & u(T-L+1) \\ \vdots & \vdots & & \vdots \\ u(L-1) & u(L) & \cdots & u(T-1) \end{bmatrix},$$

$$H_L(y_{[0,T-1]}) = \begin{bmatrix} y(0) & y(1) & \cdots & y(T-L) \\ y(1) & y(2) & \cdots & y(T-L+1) \\ \vdots & \vdots & & \vdots \\ y(L-1) & y(L) & \cdots & y(T-1) \end{bmatrix}.$$

In the remainder, we will often stack input and output Hankel matrices to obtain a single block Hankel matrix, as follows:

$$\begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix} \in M_{(m+p)L \times (T-L+1)}(\mathbb{R}), \quad (2.2)$$

which will be used in various computational tasks. It is now obvious that every column of the matrices from (2.2) is a restricted input-output trajectory. Next, definition of persistently exciting input is presented.

Definition 8. *Given an sequence of inputs $u_{[0,T-1]}$ and $k \in \{1, \dots, T\}$, we say that the sequence of inputs is persistently exciting of order k if the Hankel matrix $H_k(u_{[0,T-1]})$ is of full row rank.*

We have laid out all the necessary definitions and theorems needed for the statement of the main topic of this thesis, the Willems' Fundamental Lemma.

2.2 The Willems' Fundamental Lemma

The main result of the Lemma is that every restricted input-output trajectory of length L of the system can be obtained as a linear combination of finitely many trajectories which span the whole trajectory space, i.e., the behaviour.

Theorem 3. (*Willems' Fundamental Lemma*) *Given an LTI system (2.1), assume the pair (A, B) is controllable and let $(u_{[0,T-1]}, y_{[0,T-1]})$, $T \in \mathbb{N}$ be its restricted input-output trajectory. Let $L \in \{1, \dots, T\}$. If the input $u_{[0,T-1]}$ is persistently exciting of order $N + L$, then $(\bar{u}_{[0,L-1]}, \bar{y}_{[0,L-1]})$ is a restricted input-output trajectory of (2.1) if and only if*

$$\begin{bmatrix} \bar{u}_{[0,L-1]} \\ \bar{y}_{[0,L-1]} \end{bmatrix} = \begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix} g$$

for some $g \in \mathbb{R}^{T-L+1}$.

The original statement and proof were given by Willems et al. [1], within the behavioural framework. For completeness of presentation, we also present a proof of the Fundamental Lemma. This proof idea follows the proof given in [6], with a difference that in our proof we prove equality of subspaces slightly differently than in their proof.

Proof of Theorem 3. From recursive application of the difference equations (2.1) we obtain the following expression:

$$y(t) = CA^t x(0) + \left(\sum_{k=0}^{t-1} CA^{t-k-1} Bu(k) \right) + Du(t), \quad (2.3)$$

where $x(0) \in \mathbb{R}^n$ is the initial state. Generally, this could be adapted to any initial state $x_{ini} = x(t_{ini})$ using the time-invariance of the system. Now for the input-output trajectory $(\bar{u}_{[0,L-1]}, \bar{y}_{[0,L-1]})$ we can write the result from above in matrix notation like this:

$$\begin{bmatrix} \bar{y}(0) \\ \bar{y}(1) \\ \bar{y}(2) \\ \vdots \\ \bar{y}(L-1) \end{bmatrix} = \underbrace{\begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{L-1} \end{bmatrix}}_{\Omega_L} x(0) + \underbrace{\begin{bmatrix} D & & & \\ CB & D & & \\ CAB & CB & D & \\ \vdots & \ddots & \ddots & \ddots \\ CA^{L-2}B & \dots & \dots & D \end{bmatrix}}_{\Theta_L} \begin{bmatrix} \bar{u}(0) \\ \bar{u}(1) \\ \bar{u}(2) \\ \vdots \\ \bar{u}(L-1) \end{bmatrix}. \quad (2.4)$$

Matrix Ω_L is the classical observability matrix and Θ_L is the block Toeplitz matrix of so-called Markov parameters. We can write the expression (2.4) in a more compact form as follows:

$$\begin{bmatrix} \bar{u}_{[0,L-1]} \\ \bar{y}_{[0,L-1]} \end{bmatrix} = \begin{bmatrix} 0 & I \\ \Omega_L & \Theta_L \end{bmatrix} \begin{bmatrix} x_{ini} \\ \bar{u}_{[0,L-1]} \end{bmatrix}. \quad (2.5)$$

Since on the left side is a general input-output trajectory of length L (assuming time-invariance), we conclude that the space of all such trajectories is spanned by the columns of the $M := \begin{bmatrix} 0 & I \\ \Omega_L & \Theta_L \end{bmatrix}$. The columns of the Hankel matrix $\begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix}$ are input-output trajectories, so we conclude that:

$$\text{im} \begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix} \subseteq \text{im} \begin{bmatrix} 0 & I \\ \Omega_L & \Theta_L \end{bmatrix}.$$

To prove the Lemma we need to prove the equality of those spaces. We assume the strict inclusion:

$$\text{im} \begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix} \subsetneq \text{im} \begin{bmatrix} 0 & I \\ \Omega_L & \Theta_L \end{bmatrix}.$$

The expression (2.5) can be used to formulate an analogous expression using Hankel matrices like this:

$$\begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix} = \begin{bmatrix} 0 & I \\ \Omega_L & \Theta_L \end{bmatrix} \begin{bmatrix} X_{[0,T-L]} \\ H_L(u_{[0,T-1]}) \end{bmatrix},$$

where $X_{[0,T-L]} := [x(0) \ x(1) \ \dots \ x(T-L)]$.

By our assumption the matrix $J_L := \begin{bmatrix} X_{[0,T-L]} \\ H_L(u_{[0,T-1]}) \end{bmatrix}$ must not have full row rank. If it were full row rank, then the rank of Hankel matrix on the left would be equal to the rank of J_L and thus the image spaces would be equal. That implies that the left kernel of M is non-trivial, in other words, there exists a non-zero vector $v = (a^T, b_0^T, b_1^T, \dots, b_{L-1}^T)$, $a \in \mathbb{R}^n$, $b_i \in \mathbb{R}^m$ such that $vJ_L = 0$. Expanding we get:

$$a^T x(t) + \sum_{k=0}^{L-1} b_k^T u(t+k) = 0, \quad \forall t \in \{0, 1, \dots, T-L\}. \quad (2.6)$$

Shifting the above equation by an arbitrary $j \in \mathbb{N}$ and solving the difference equations of (2.1) for the shifted state (similar to (2.5)) we obtain:

$$a^T A^j x(t) + \sum_{i=0}^{j-1} a^T A^{j-i-1} B u(t+i) + \sum_{k=0}^{L-1} b_k^T u(t+k+j) = 0. \quad (2.7)$$

We want to remove the state $x(t)$ from the equation. The easiest way to do it is to leverage the Cayley-Hamilton theorem, which states that matrix A satisfies its characteristic polynomial $p(\lambda) = \lambda^n + \alpha_{n-1}\lambda^{n-1} + \dots + \alpha_1\lambda + \alpha_0$;

$$p(A) = A^n + \alpha_{n-1}A^{n-1} + \dots + \alpha_1A + \alpha_0I = 0$$

Now, if we take $n+1$ copies of equation (2.7) for each $j = 0, 1, \dots, n$, and multiply each one with its corresponding α_j and sum them all, the terms with $x(t)$ cancel as a result of C-H theorem. Now we get the following expression with only the inputs:

$$\sum_{l=0}^{n+L-1} \beta_l u(t+l) = 0, \quad \forall t \in \{0, 1, \dots, T-L\}. \quad (2.8)$$

The expression for β_l -s can be computed by equating terms with the sum in which C-H was used and the result is:

$$\beta_l := \sum_{j=\max(0, l-L+1)}^{\min(n, l)} \alpha_j b_{l-j}^T + a^T \sum_{j=l+1}^n \alpha_j A^{j-l-1} B. \quad (2.9)$$

Equation (2.8), by the definition of a Hankel matrix, can be rewritten as:

$$\begin{bmatrix} \beta_0 & \beta_1 & \dots & \beta_{n+L-1} \end{bmatrix} H_{n+L}(u_{[0, T-1]}) = 0.$$

Since $u_{[0, T-1]}$ is persistently exciting of order $N + L$, and $N \geq n$, it is clear that $H_{n+L}(u_{[0, T-1]})$ is of full row rank, so we conclude that:

$$\beta_0 = \beta_1 = \dots = \beta_{n+L-1} = 0.$$

From (2.9) if $l \geq n$ the right sum is empty so $\beta_n, \dots, \beta_{n+L-1}$ depend only on b_k -s. Now we get:

$$\beta_{n+L-1} = \alpha_n b_{L-1}^T \underbrace{=}_{\alpha_n = 1} b_{L-1}^T = 0,$$

$$\beta_{n+L-2} = b_{L-2}^T + \alpha_{n-1} b_{L-1}^T = b_{L-2}^T = 0,$$

continuing until $l = n$, we get $b_0 = b_1 = \dots = b_{L-1} = 0$; so, the left sum of (2.9) is 0, and thus, for $l \leq n$, the right sum remains. Now:

$$\beta_{n-1} = \alpha_n a^T B = a^T B = 0,$$

$$\beta_{n-2} = a^T (AB + \alpha_{n-1} B) = a^T AB + a^T \alpha_{n-1} B \implies a^T AB = 0.$$

continuing until $l = 0$ we obtain the following result:

$$a^T \underbrace{\begin{bmatrix} B & AB & \dots & A^{n-1} B \end{bmatrix}}_{\mathcal{C}} = 0.$$

Since (A, B) is controllable then $\mathcal{C}$ is of full rank, and thus $a = 0 \implies v = 0$, which contradicts our assumption. Thus:

$$\text{im} \begin{bmatrix} H_L(u_{[0, T-1]}) \\ H_L(y_{[0, T-1]}) \end{bmatrix} = \text{im} \begin{bmatrix} 0 & I \\ \Omega_L & \Theta_L \end{bmatrix},$$

which means there exists a $g \in \mathbb{R}^{T-L+1}$ such that for any restricted input-output trajectory $(\bar{u}_{[0, L-1]}, \bar{y}_{[0, L-1]})$ we have:

$$\begin{bmatrix} \bar{u}_{[0, L-1]} \\ \bar{y}_{[0, L-1]} \end{bmatrix} = \begin{bmatrix} H_L(u_{[0, T-1]}) \\ H_L(y_{[0, T-1]}) \end{bmatrix} g.$$

The proof above is just the only if direction; however, since each column of the stacked Hankel matrix is an restricted input-output trajectory, the if direction follows directly from that. □

Stability of the system is not needed because the Lemma uses finite-horizon trajectories and thus doesn't require boundedness of state as $t \rightarrow \infty$. Also, the observability is not needed as it can be seen in the formulation and proof. In the proof we have also showed that the matrix:

$$\begin{bmatrix} X_{[0,T-L]} \\ H_L(u_{[0,T-1]}) \end{bmatrix}$$

is of full rank. Also we have made an indirect assumption regarding row rank for persistence of excitation. For a matrix to have full row rank, a necessary condition is that the number of columns needs to be at least equal to the number of rows, thus the persistence of excitation condition needs that condition fulfilled. When we impose that condition on $H_{N+L}(u_{[0,T-1]})$ we get that:

$$T - (N + L + 1) \geq m(N + L) \iff T \geq (m + 1)(N + L) - 1. \quad (2.10)$$

So for persistence of excitation the length of initial trajectory T needs to satisfy the condition above for a given L . Generally, for a long enough recorded trajectory we can use the Lemma for a shorter length trajectory. In our Lemma formulation we only used trajectories of length L starting from 0, but it can be shown that the Lemma holds for a general restricted input-output trajectory of length L , i.e., for time window of $[i, i + L - 1] \forall i \in \mathbb{Z}_{\geq 0}$ we have:

$$\begin{bmatrix} \bar{u}_{[i,i+L-1]} \\ \bar{y}_{[i,i+L-1]} \end{bmatrix} = \begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix} g. \quad (2.11)$$

Now we will define the lag ℓ of the system: smallest positive integer k s.t. $\text{rank } \Omega_k = \text{rank } \Omega_{k+1}$, where Ω_k is the observability matrix from (2.5), is called the lag of the system; in other words, the smallest k after which the rank of the observability matrix does not change. Clearly, by C-H theorem, $\ell \leq n$. For an observable system that is the minimal trajectory length such that the initial state x_{ini} can be reconstructed. It can also be shown that $\text{rank}(M) = \text{rank}(\Omega_L) + mL$, where m is the number of inputs. Since in the right column of M there is an identity matrix and on the left is the zero matrix, than those right columns are linearly independent and none of the left columns can be gotten as a linear combination of right columns; thus, the rank of M is the rank of right columns - mL added with the rank of left columns which is the rank of Ω_L .

2.3 Lemma - Numerical Examples

In this chapter we will demonstrate the Fundamental Lemma on several numerical examples. We will use the system model to gather the necessary data for the Hankel matrices and then use them to verify for a random trajectory that it lies in the image of the Hankel matrix. For this we use MATLAB [11] because it is easier to work with its Control System Toolbox for system simulation.

2.3.1 Simple SISO System

Consider a discrete-time LTI system (2.1) with state-space matrices:

$$A = \begin{bmatrix} 0.5 & 0.1 & 0.1 \\ 0.1 & 0.4 & 0.2 \\ -0.1 & 0.2 & 0.3 \end{bmatrix} \quad B = \begin{bmatrix} -1 \\ 0.1 \\ 0.7 \end{bmatrix} \quad (2.12)$$

$$C = \begin{bmatrix} 0.1 & 2 & 1.5 \end{bmatrix} \quad D = \begin{bmatrix} 0 \end{bmatrix} \quad (2.13)$$

Clearly $n = 3, m = p = 1$ and the eigenvalues of A are $\{0.5834, 0.4657, 0.1509\}$ which all have their absolute value less than 1 which means the system is stable. Also the controllability and the observability matrices, which are in this case square matrices, have a non-zero determinant. The system is therefore both controllable and observable. The length of initial trajectory and length of trajectories for reconstruction are chosen respectively as $T = 50$ and $L = 22$. It can be verified that for $N = n$ the condition (2.10) is satisfied. For the input for this example we are using a randomly generated sequence obtained by using MATLAB's *rand* function. Since the input is random there is a high chance for persistency of excitation. Before testing the lemma we check the rank of the $H_{N+L}(u_{[0,T-1]})$. We input the random input sequence to the system and record the output data. We then create the Hankel matrix $\begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix}$. Now we generate a new random input-output trajectory of the system $(\bar{u}_{[0,L-1]}, \bar{y}_{[0,L-1]})$. We now use the Hankel matrix and the new data to find the vector g . We impose that as a least-squares problem since the Hankel matrix is generally not square and the system of equations is underdetermined (fewer rows than columns). The objective for a minimization problem is the following:

$$\text{Find } g \in \mathbb{R}^{T-L-1} \text{ that minimizes } \left\| \begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix} g - \begin{bmatrix} \bar{u}_{[0,L-1]} \\ \bar{y}_{[0,L-1]} \end{bmatrix} \right\|_2^2. \quad (2.14)$$

In MATLAB we solve the following problem using the *lsqminnorm* function, which uses *complete orthogonal decomposition (COD)* instead of *singular value decomposition (SVD)* to compute g . However, for our example both SVD and COD should yield similar results. Now the Lemma states that g should make that norm of the difference equal to zero. That means the g is an exact solution, not an approximation. Computing the norm for the calculated g we get result in MATLAB:

$$\left\| \begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix} g - \begin{bmatrix} \bar{u}_{[0,L-1]} \\ \bar{y}_{[0,L-1]} \end{bmatrix} \right\| \leq 10^{-10}. \quad (2.15)$$

The norm is not exactly 0, but it is really close. The small error arises from truncation of real numbers due to floating-point arithmetic, so we can assume the result is valid. Our reconstructed trajectory is found in the image of the Hankel matrix created from recorded data.

2.3.2 Mechanical MIMO System

In this example we will demonstrate the Lemma on a concrete mechanical system. We will show, like in the last example, that we can reconstruct a trajectory from previously recorded data.

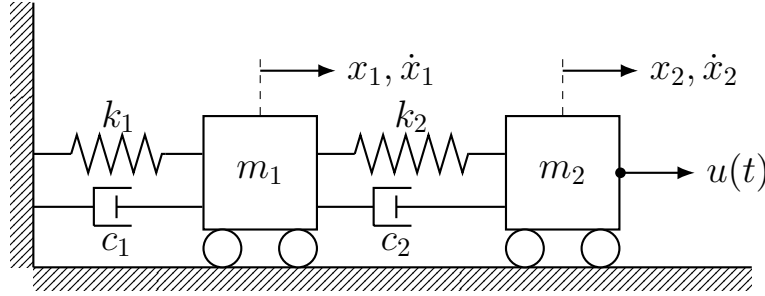


Figure 2.1: Double spring-mass-damper system with force input acting on the second mass.

The states of the system are positions and velocities of each mass. The input is a force on the second mass. We chose the output to be a vector of positions of both masses; thus, this is a MIMO system. Using mechanical equilibrium equations we can get a system of coupled linear differential equations which can be represented as a continuous state-space LTI system:

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t), \\ y(t) &= Cx(t) + Du(t),\end{aligned}$$

where

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_1+k_2}{m_1} & -\frac{c_1+c_2}{m_1} & \frac{k_2}{m_1} & \frac{c_2}{m_1} \\ 0 & 0 & 0 & 1 \\ \frac{k_2}{m_2} & \frac{c_2}{m_2} & -\frac{k_2}{m_2} & -\frac{c_2}{m_2} \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_2} \end{bmatrix} \quad (2.16)$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (2.17)$$

For this example the parameters are: $m_1 = 1.0 \text{ kg}$, $m_2 = 0.8 \text{ kg}$, $k_1 = 120 \text{ Nm}^{-1}$, $k_2 = 180 \text{ Nm}^{-1}$, $c_1 = 1.5 \text{ Nsm}^{-1}$, $c_2 = 2.0 \text{ Nsm}^{-1}$. The above presented is a continuous state-space model, but we need a discrete model so we need to discretize it. There are many ways of discretizing the continuous model. We will derive the discrete version matrices A_d, B_d, C_d, D_d using zero-order hold (ZOH). Assume sampling period T_s , we chose $T_s = 10^{-2} \text{ s}$. Now we will begin from the solution of the general continuous state-space system:

$$x(t) = e^{A(t-t_0)}x(t_0) + \int_{t_0}^t e^{A(t-\tau)}Bu(\tau)d\tau.$$

Let $k \in \mathbb{Z}_{\geq 0}$ and $t_0 = kT_s$, $t = (k+1)T_s$. Define $x_k := x(kT_s)$, $u_k := u(kT_s)$. Now we have:

$$x_{k+1} = e^{AT_s}x_k + \int_{kT_s}^{(k+1)T_s} e^{A[(k+1)T_s-\tau]}Bu(\tau)d\tau.$$

Now assume $u(\tau) = u(kT_s) = u_k = \text{const.}$ $\forall \tau \in (kT_s, (k+1)T_s)$, $\forall k \in \mathbb{Z}_{\geq 0}$. This is the ZOH. We assume the input is constant between sampling. After a change of variables for integration, we arrive at this result. Since B and u_k are constants, they can go outside the integral.

$$x_{k+1} = e^{AT_s}x_k + \left(\int_0^{T_s} e^{Az}dz \right) Bu_k,$$

$$y_k = Cx_k + Du_k.$$

We can now see that this is essentially (2.1) but where:

$$A_d = e^{AT_s}, \quad B_d = \left(\int_0^{T_s} e^{Az}dz \right) B, \quad C_d = C, \quad D_d = D.$$

This was a general derivation. In MATLAB we use the function `c2d` to convert the system from continuous to discrete time. Now we can see that $n = 4$, $m = 1$, $p = 2$. The initial trajectory we recorded is 30 seconds long, in MATLAB this was converted to number of samples with $T = \text{round}(30/T_s)$. The length of reconstructed trajectories L is 12 seconds. We set $N = n$ because the systems has four energy tanks. For the input, instead of a random sequence of real numbers between 0 and 1, we use a pseudo-random binary sequence (PRBS) - a binary signal which jumps randomly between +1 and -1. We multiply that signal with a positive real number (amplitude) and feed it through a low-pass filter to make it a bit smoother and more realistic. This choice of T , L and N satisfy the (2.10).

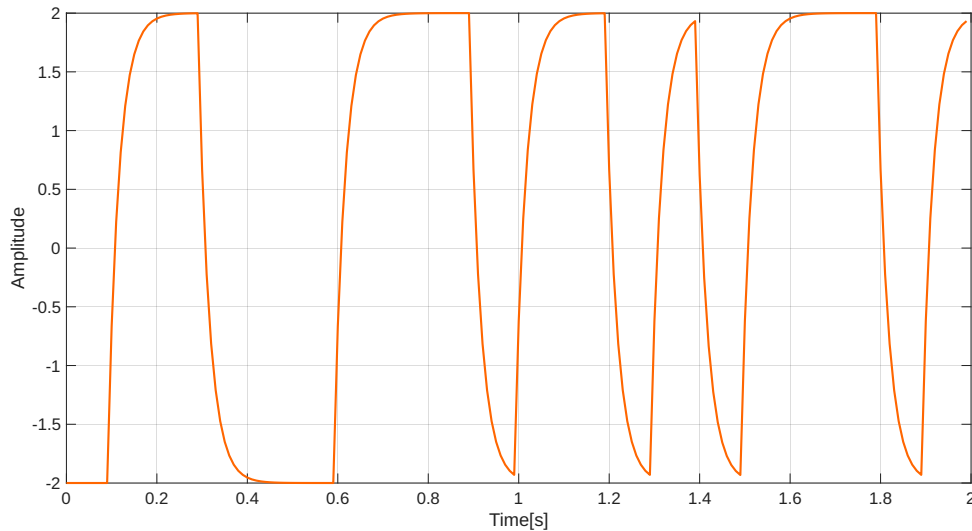
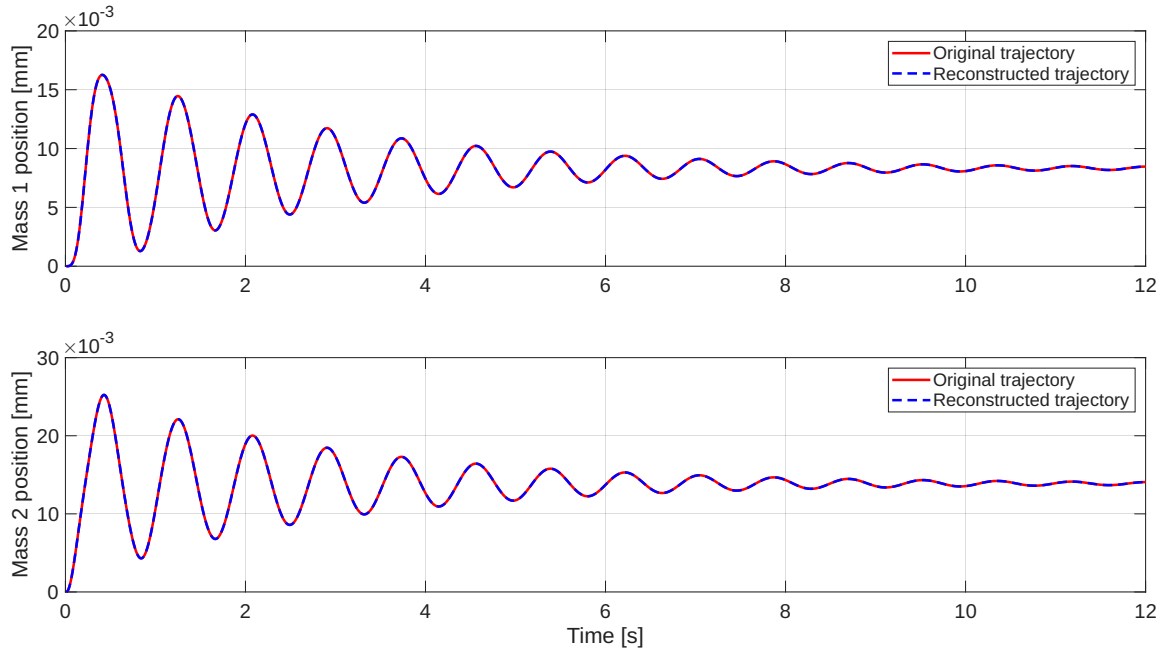


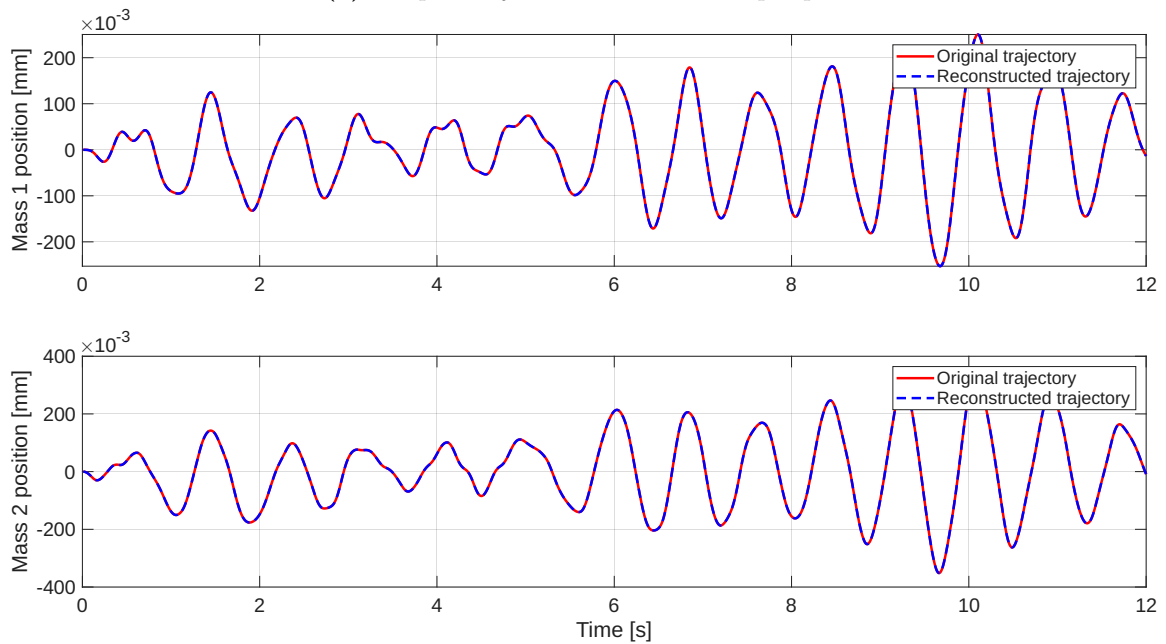
Figure 2.2: Low-pass filtered PRBS input signal

Since this type of input data has enough "randomness" it is a good choice for a persistently exciting input. And after testing the rank condition for persistence of excitation it was always satisfied.

Now, as before, we input that persistently exciting signal to the system and place the recorded trajectories into the corresponding Hankel matrix. We will test the Lemma on two inputs: first will be a step function, and second will be a new PRBS signal. We simulate the output trajectories using the system model and solve for vector g like in (2.14).



(a) Output trajectories for a unit step input



(b) Output trajectories for a PRBS input

Figure 2.3: Comparison of original and reconstructed trajectories of the system

It can be seen from Figure 2.3 that trajectories line up. This is confirmed numerically with (2.15) seeing that for both trajectories we have the norm $\leq 10^{-10}$. Thus, the Lemma holds for this example.

3 Data-driven simulation of LTI systems

In the last section we used the Fundamental lemma to demonstrate that any sufficiently long trajectory of length L can be found in the image of a Hankel matrix that is built using a previous trajectory with persistently exciting input. This is really interesting, however, there are no direct use cases when it comes to control. In this section we will build up on the Fundamental lemma and use it to simulate the future of a given trajectory using the same Hankel matrix.

3.1 Problem statement and solution

Consider the system (2.1) and assume the pair (A, B) is controllable. Let $L_{ini} \in \mathbb{Z}_{\geq 0}$ be the length of the past trajectory and suppose we want to simulate the future trajectory of length $L_{ref} \in \mathbb{Z}_{\geq 0}$. Let $L := L_{ini} + L_{ref}$. Assume the following is known:

- upper bound N on the state dimension, i.e., $N \geq n$,
- restricted input-output trajectory $(u_{[0,T-1]}, y_{[0,T-1]})$, where T satisfies (2.10),
- an initial restricted input-output trajectory $(\bar{u}_{[0,L_{ini}-1]}, \bar{y}_{[0,L_{ini}-1]})$,
- reference input $\bar{u}_{[L_{ini},L-1]}$.

We want to calculate the output $\bar{y}_{[L_{ini},L-1]}$ such that $(\bar{u}_{[0,L-1]}, \bar{y}_{[0,L-1]})$ is a restricted input-output trajectory of (2.1). To summarize, based on the recorded data, the previous history of a trajectory and future input we want to find the future output part of that trajectory. We can split the Hankel matrices of recorded data like this:

$$\begin{bmatrix} H_L(u_{[0,T-1]}) \\ H_L(y_{[0,T-1]}) \end{bmatrix} = \begin{bmatrix} U_p \\ U_f \\ Y_p \\ Y_f \end{bmatrix}, \quad (3.1)$$

where U_p and U_f have mL_{ini} and mL_{ref} and Y_p and Y_f have pL_{ini} and pL_{ref} rows, respectively. Next we present a solution of the above described problem, as it is done in [6].

Theorem 4. *Consider a system (2.1) with (A, B) controllable and persistently exciting input $u_{[0,T-1]}$ of order $N+L$. Suppose restricted input-output trajectory $(\bar{u}_{[0,L_{ini}-1]}, \bar{y}_{[0,L_{ini}-1]})$ and reference input $\bar{u}_{[L_{ini},L-1]}$ are known. Then the following statements hold:*

(i) *There exists $g \in \mathbb{R}^{T-L+1}$ such that:*

$$\begin{bmatrix} U_p \\ Y_p \\ U_f \end{bmatrix} g = \begin{bmatrix} \bar{u}_{[0,L_{ini}-1]} \\ \bar{y}_{[0,L_{ini}-1]} \\ \bar{u}_{[L_{ini},L-1]} \end{bmatrix}.$$

(ii) *If g is a solution to (i) then define the output $\bar{y}_{[L_{ini},L-1]} = Y_f g$. Such defined output makes $(\bar{u}_{[0,L-1]}, \bar{y}_{[0,L-1]})$ a restricted input-output trajectory of (2.1).*

(iii) *If $L_{ini} \geq \ell$ (the lag of the system 2.2) then such output in (ii) is the only one that makes $(\bar{u}_{[0,L-1]}, \bar{y}_{[0,L-1]})$ a restricted input-output trajectory, i.e., it is unique.*

3.2 Simulating LTI system from data

Theorem 4 in fact presents the method of how to simulate the future trajectory of the system, based on the past data. We will test this result on numerical examples. Firstly, we will future output trajectory for the LTI MIMO example from the previous chapter (2.1). We simulate the system with a PRBS input for 50 seconds ($T_s = 0.01$ s, $T = 5000$) and recorded the trajectory ($u_{[0,T-1]}, y_{[0,T-1]}$). We chose the following lengths of past and future (simulated) part of the trajectory:

$$L_{ini} = 500, L_{ref} = 1000, L = L_{ini} + L_{ref} = 1500.$$

We populated the $H_L(u_{[0,T-1]})$ and $H_L(y_{[0,T-1]})$ and split them into (U_p, Y_p, U_f, Y_f) parts. Result from (4, (i)) tells us g exists and we calculate it using the modified least-squares problem from the previous section (2.14):

$$\text{Find } g \in \mathbb{R}^{T-L-1} \text{ that minimizes } \left\| \begin{bmatrix} U_p \\ Y_p \\ U_f \end{bmatrix} g - \begin{bmatrix} \bar{u}_{[0,L_{ini}-1]} \\ \bar{y}_{[0,L_{ini}-1]} \\ \bar{u}_{[L_{ini},L-1]} \end{bmatrix} \right\|_2^2. \quad (3.2)$$

Solving with MATLAB's *lsqminnorm*, we obtain the vector g . Using g we calculate the future output trajectory $\bar{y}_{[L_{ini},L-1]} = Y_f g$. By theorem (4) this output part is unique since $L_{ini} \geq \ell = 2$ (for the parameters given in (2.16)); generally, the lag is unknown, but if we take large enough L_{ini} we can be sure that it is larger. Next, we compare the data-based future output trajectories with one calculated using the system model — the first L_{ini} points using the model. For the input signal we used a multi-sine input like this:

$$u(k) = 10 \sin(2\pi k T_s) + \frac{1}{3} \sin(2\pi 3k T_s), \quad k \in \{0, 1, \dots, L-1\}.$$

The simulation output presented in Figure 3.1:

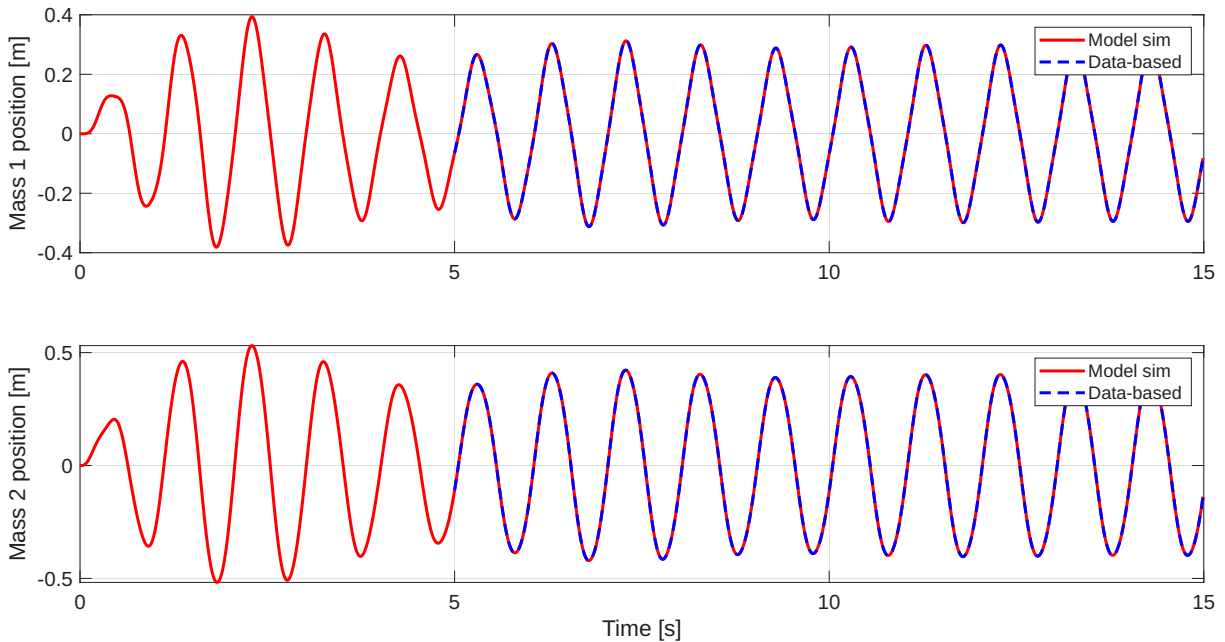


Figure 3.1: Data-based vs. model-based future trajectory simulation

We can see that the data-based simulated trajectories perfectly match the model-based ones. Numerically, like before, we get the following confirmation that the future output trajectories coincide:

$$\left\| \bar{y}_{[L_{ini}, L-1]} - \bar{y}_{[L_{ini}, L-1]}^{model} \right\| \leq 10^{-10}.$$

3.3 Simulating a nonlinear system around equilibrium

In the last section we showed that it is possible to simulate a LTI system directly from recorded data, without a model. For nonlinear systems, a classical approach is to linearize the dynamics around a stationary point around which the linearized model is mostly valid. We want to gather the data around the stationary (equilibrium) point of the system and apply the data-based simulation of the future trajectory. For this example we will take a classical torque-driven pendulum:

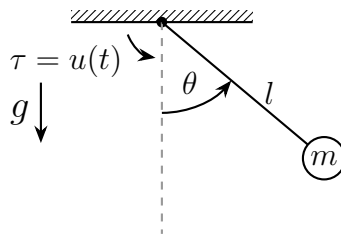


Figure 3.2: Torque-driven simple pendulum

Parameters for the pendulum used in simulations are:

$$m = 0.2 \text{ kg}, \quad l = 0.5 \text{ m}, \quad b = 0.05 \text{ Nms}, \quad g = 9.81 \frac{\text{m}}{\text{s}^2}.$$

Assuming damping is proportional to angular velocity with proportionality constant b , we write the state-space dynamics with the state vector $x = (\theta, \dot{\theta})^T$ as follows:

$$\dot{x} = \frac{d}{dt} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{\theta} \\ \frac{-b}{ml^2} \dot{\theta} - \frac{g}{l} \sin(\theta) + \frac{u}{ml^2} \end{bmatrix} = f(x, u). \quad (3.3)$$

Since this is in continuous state-space formulation we need to derive a discrete-time version of it. Using sampling period $T_s = 0.01 \text{ s}$ and assuming ZOH on the input $u(t)$ (see 2.3.2) we discretize (3.3) using the Runge-Kutta 4 [12] method as follows:

$$x_{k+1} = x_k + \frac{T_s}{6} (k_1 + 2k_2 + 2k_3 + k_4), \quad (3.4)$$

where

$$\begin{aligned} k_1 &= f(x_k, u_k), \\ k_2 &= f\left(x_k + \frac{T_s}{2} k_1, u_k\right), \\ k_3 &= f\left(x_k + \frac{T_s}{2} k_2, u_k\right), \\ k_4 &= f(x_k + T_s k_3, u_k). \end{aligned}$$

Assuming initial state $x_0 = (0, 0)^T$ and that the output is the full state of the system ($y_k = x_k$), we supply a pseudo-random binary sequence input for 50 seconds to the system and record the output. We configured the amplitude of the PRBS such that angle θ is always $\leq \frac{\pi}{18}$ (10 degrees).

Since θ is small enough and the model is approximately linear around 0, we will try to simulate the future trajectory of the system using the Fundamental lemma. We chose the following trajectory lengths $L_{ini} = 50$, $L_{ref} = 600 \implies L = 650$. We assumed $N = 4$. Since a PRBS input is persistently exciting of order $N + L$ we can repeat the procedure in 3.2. We give a zero input and set the initial state to $\left(\frac{\pi}{10}, 0\right)^T$ for the simulating trajectory - autonomous system.

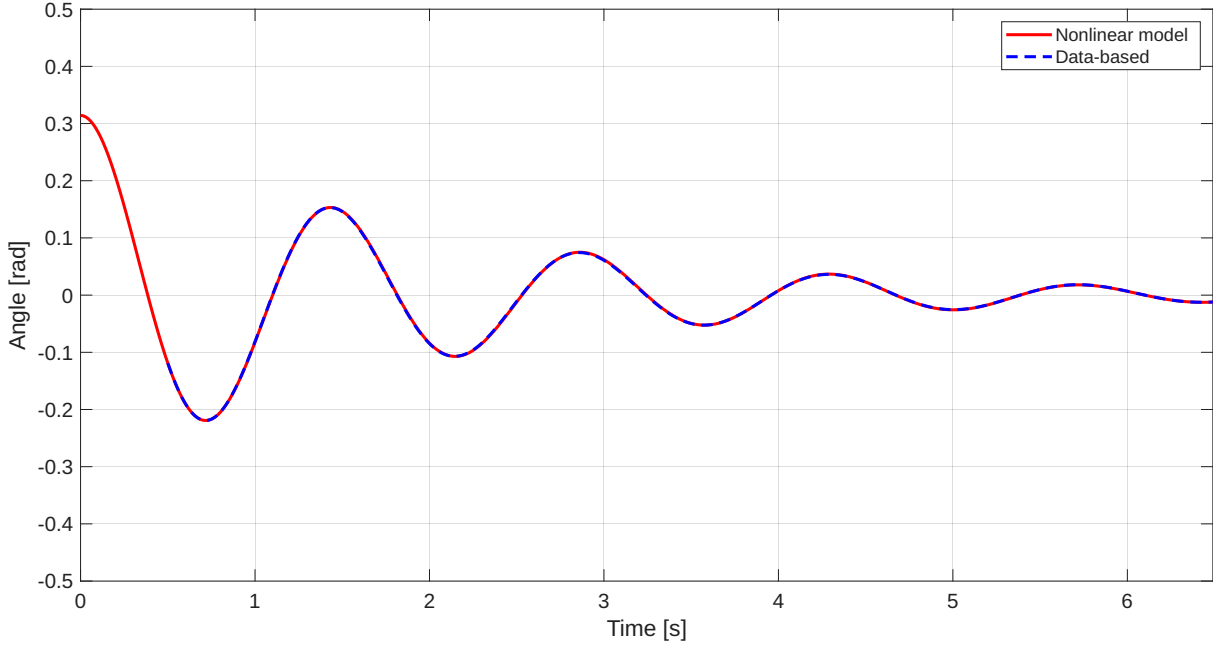


Figure 3.3: Pendulum trajectory simulation - data-based vs. model

If we now calculate the norm of the difference between the two output trajectories we get:

$$\left\| \bar{y}_{[L_{ini}, L-1]}^{data} - \bar{y}_{[L_{ini}, L-1]}^{model} \right\| \sim 10^{-2}.$$

Our trajectories do not coincide exactly, but that is to be expected because the model is nonlinear. Since the error is small, we can say that data-based simulation works well for small angles. Now we will try simulating with a larger angle and we changed $L_{ini} = 10$. We will also compare a linearized model of (3.3) which uses $\sin(\theta) \approx \theta$ linearization. For initial angle of $\frac{\pi}{3}$ and zero input we get the following results (for a specific simulation):

$$\left\| \bar{y}_{[L_{ini}, L-1]}^{data} - \bar{y}_{[L_{ini}, L-1]}^{model} \right\| \approx 0.428618, \quad (3.5)$$

$$\left\| \bar{y}_{[L_{ini}, L-1]}^{lin.model} - \bar{y}_{[L_{ini}, L-1]}^{model} \right\| \approx 1.446429. \quad (3.6)$$

We can see that the error for the data-based simulation is quite smaller, which can be seen on the Figure 3.4:

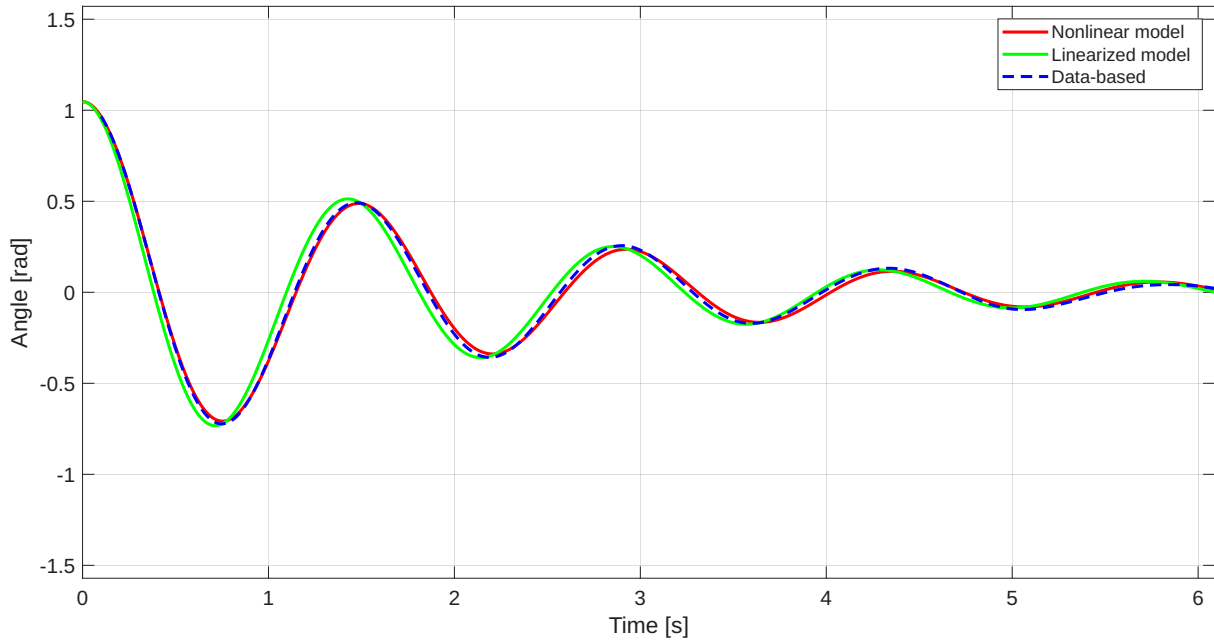


Figure 3.4: Pendulum trajectory simulation - large angles; - data-based vs. model vs. linearized model

We can see that the data-based simulation is closer to the nonlinear model one and it is considerably more accurate for larger angles. However, this MATLAB script inputs a new PRBS to the system for the data every time the script is run. Randomness of that signal implies that the result from above is not general. In fact, we sometimes got that the linearized model produces a lower error than the data-based. In practice, this could be fixed by recording multiple data sets with PRBS and keeping the one which makes the data-based simulation the best. We ran the MATLAB script 1000 times and recorded the norm of the difference for each one. The linearized model error (3.6) is constant because it does not depend on a PRBS input. These are the results:

$$\begin{aligned} \left\| \bar{y}_{[L_{ini}, L-1]}^{data} - \bar{y}_{[L_{ini}, L-1]}^{model} \right\|_{average} &= 0.685848, \\ \left\| \bar{y}_{[L_{ini}, L-1]}^{data} - \bar{y}_{[L_{ini}, L-1]}^{model} \right\|_{max} &= 3.47226, \\ \left\| \bar{y}_{[L_{ini}, L-1]}^{data} - \bar{y}_{[L_{ini}, L-1]}^{model} \right\|_{min} &= 0.238929. \end{aligned}$$

This illustrated that in some cases with data-based simulation we can get a better result than a linearized model, however, if we do not supply a "good" persistently exciting input we can get worse results.

In this chapter we demonstrated that it is possible to simulate future trajectories of LTI systems directly from recorded data and knowledge about the input. This hints at the following chapter which uses this approach to implement predictive control of such systems entirely from input-output data.

4 Data-Enabled Predictive Control (DeePC)

4.1 Introduction

Classical control theory almost always uses a model of the system to synthesise a controller. Usually that model is obtained by system identification from recorded data, but when the model is too complex to be easily identified or approximated from first principles people started developing control methods based solely on data. During the last decade, data availability and processing power have been growing rapidly; thus, data-driven control methods have started to develop intensively. There has also been a lot of progress in the machine learning theory. In this chapter we will build up on results from the previous sections to create a controller that uses only input/output data, completely bypassing a model. A needed prerequisite for our data-based controller is model predictive control (MPC) [13], an optimal control algorithm which finds optimal input to the system on a finite and receding horizon whose data is constrained with the model and other (physical) limitations. The DeePC algorithm, essentially, does the same thing as MPC - same optimization problem - just instead of the model it uses previously recorded data for the prediction part. This algorithm relies on the Fundamental Lemma since its result states that it is possible to predict future trajectory of a linear system given input data and recorded data. First formulation (and naming) of DeePC was given by [3]. Illustration of data-based and model-based control design paradigms can be seen in Figure 4.1. Also, illustration of predictive control can be seen in Figure 4.2.

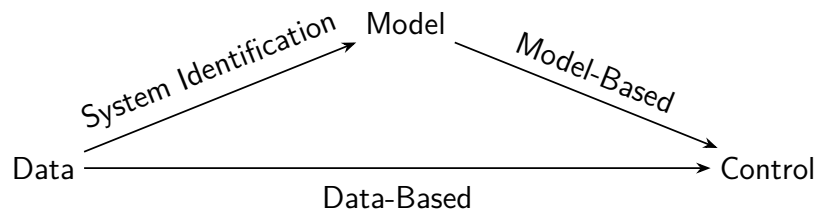


Figure 4.1: Different control design approaches from data

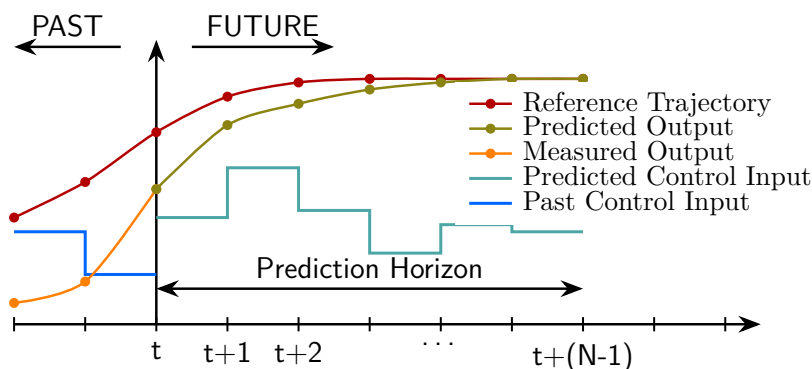


Figure 4.2: General predictive control illustration

4.2 Model predictive control

As mentioned before, MPC is a control algorithm which uses optimization with system model and limitations as constraints to find optimal control input over a finite, receding horizon. At each time step, future trajectory of the system is predicted using the model and optimal sequence of control input is computed taking constraints on inputs and states of the system. Typically, only the first input of the optimal input sequence is applied, after which the procedure is repeated with new measurements. We can say that, other than MPC, we do not have another control solution for multivariable systems with constraints on inputs and outputs with stability guarantees. In this section we will formulate classical MPC for LTI systems which is needed in the subsequent section for formulation and understanding of DeePC.

Given an LTI system (2.1), the classical finite-horizon MPC optimization problem at time step t can be written as:

$$\begin{aligned} \min_{x,y,u} \quad & \sum_{j=0}^{N-1} \left(\|y_j - r_{t+j}\|_Q^2 + \|u_j\|_R^2 \right) \\ \text{subject to: } \quad & x_0 = \hat{x}(t), \\ & x_{j+1} = Ax_j + Bu_j, \quad j = 0, \dots, N-1; \\ & y_j = Cx_j + Du_j, \quad j = 0, \dots, N-1; \\ & y_j \in \mathcal{Y}, \quad j = 0, \dots, N-1; \\ & u_j \in \mathcal{U}, \quad j = 0, \dots, N-1; \end{aligned} \tag{4.1}$$

Where: $\|z\|_P^2 := z^T P z$, for a symmetric positive semidefinite matrix P (notation $P \succeq 0$). Matrix $Q \in M_{p \times p}(\mathbb{R})$, $Q \succeq 0$ is the output cost matrix and $R \in M_{m \times m}(\mathbb{R})$, $R \succ 0$ is the positive definite input cost matrix. Sets $\mathcal{U} \subseteq \mathbb{R}^m$, $\mathcal{Y} \subseteq \mathbb{R}^p$ are input and output constraints set, respectively. Sequence $(r_0, r_1, \dots)$ is the reference trajectory which we want to track, $N \in \mathbb{Z}_{>0}$ is the prediction horizon length. We use estimated state $\hat{x}$ instead of the true state because, generally, the output is not the whole state, i.e., we cannot measure the whole state; thus, we need to estimate it. This MPC optimization problem is usually a convex quadratic program (QP) [14]. Now we will lay out the MPC algorithm for reference tracking.

Algorithm 1 Model predictive control (MPC)

Require: System model (A, B, C, D) ; reference trajectory r ; past input/output data (u, y) ; constraints sets $\mathcal{U}, \mathcal{Y}$; cost matrices Q, R ;
for $t = 0, 1, \dots$ **do**
 [1] Estimate $\hat{x}(t)$ from previous input/output data
 [2] Solve (4.1) for $u^* = (u_0^*, u_1^*, \dots, u_{N-1}^*)$
 [3] Apply input $u(t) = u_0^*$
 [4] Update past input/output data with new inputs/measurements
end for

This formulation of MPC assumes stationary state inputs are 0; also, terminal cost is omitted because when we transfer to DeePC we do not know the full state of the system. It can be shown, under typical assumptions, that MPC algorithm is recursively feasible

and stabilizing [15]. This algorithm can produce excellent results if the system model is precise and if it is possible to estimate the initial state $\hat{x}(t)$.

4.3 DeePC for LTI systems

In this section we will formulate predictive control based on data. We will use the same simulation method for prediction from Chapter 3. The following are required for the DeePC algorithm:

- (i) an upper bound on state dimension N ,
- (ii) past/future trajectory lengths L_{ini} and L_{ref} , with $L_{ini} \geq \ell$; $L := L_{ini} + L_{ref}$,
- (iii) recorded restricted input-output trajectory $(u_{[0,T-1]}, y_{[0,T-1]})$, where $u_{[0,T-1]}$ is persistently exciting of order $N+L$ and $T \geq (m+1)(N+L) - 1$; m is the number of inputs,
- (iv) initial (past) trajectory $(\bar{u}_{ini}, \bar{y}_{ini}) = (\bar{u}_{[0,L_{ini}-1]}, \bar{y}_{[0,L_{ini}-1]})$,
- (v) reference trajectory $(r_0, r_1, \dots) \in (\mathbb{R}^p)^{\mathbb{Z}_{\geq 0}}$,
- (vi) input and output cost matrices $Q \succeq 0$, $R \succeq 0$,
- (vii) constraints sets $\mathcal{U}$, $\mathcal{Y}$,

Requirement (ii) assumes initial trajectory length longer than the lag of the system which is typically unknown. In practice we can simply take long enough initial trajectory for which we can be certain that it is longer than the lag. That condition is essentially equivalent to having long past input-output data for initial state estimation in MPC. Now we define matrices U_p, U_f, Y_p, Y_f like in (3.1). Conditions above allow for application of Theorem 4. We now formulate the DeePC optimization problem at time step t :

$$\begin{aligned}
 & \min_{g, y, u} \sum_{j=0}^{L_{ref}-1} \left(\|\bar{y}_j - r_{t+j}\|_Q^2 + \|\bar{u}_j\|_R^2 \right) \\
 & \text{subject to: } \begin{bmatrix} U_p \\ Y_p \\ U_f \\ Y_f \end{bmatrix} g = \begin{bmatrix} \bar{u}_{ini} \\ \bar{y}_{ini} \\ \bar{u} \\ \bar{y} \end{bmatrix} \\
 & \bar{y}_j \in \mathcal{Y}, \quad j = 0, \dots, L_{ref} - 1, \\
 & \bar{u}_j \in \mathcal{U}, \quad j = 0, \dots, L_{ref} - 1,
 \end{aligned} \tag{4.2}$$

where:

$$\begin{aligned}
 \bar{u} &= \text{col}(\bar{u}_0, \dots, \bar{u}_{L_{ref}-1}), \\
 \bar{y} &= \text{col}(\bar{y}_0, \dots, \bar{y}_{L_{ref}-1}).
 \end{aligned}$$

We deviate from previous nomenclature in favor of easier formulation. Initial trajectory (u_{ini}, y_{ini}) will be updated with new measured data such that it keeps the same length, thus the oldest data pair will be removed.

We can see that the DeePC optimization problem is *de facto* a MPC problem but where instead of prediction using the model we use previous input-output data, thus this problem avoids the model completely. We now formulate the DeePC algorithm:

Algorithm 2 Data-Enabled predictive control (DeePC)

Require: Requirements (i) - (vii) listed above

```

for  $t = 0, 1, \dots$  do
  [1] Solve (4.2) for the vector  $g$ 
  [2] Compute optimal input sequence  $\bar{u}^* = U_f g$ 
  [3] Apply the first input  $u(t) = \bar{u}^*(0)$ 
  [4] Update the initial trajectory  $(u_{ini}, y_{ini})$  with new inputs/measurements
end for
  
```

It can be shown that this DeePC is equivalent to MPC under the assumptions of the Fundamental Lemma, i.e., given the same reference and initial trajectory, they will give the identical output.

4.3.1 Exact LTI system example

Now, will we try to control the double spring-mass-damper system 2.1 with DeePC. We will try to control the position of mass 2 so we changed the output equation ($C = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}$, $D = 0$) to only give position of mass 2 to reduce computational cost. We put an input constraint $-100 \leq u(t) \leq 100$. We chose the following starting parameters $T = 100$, $L_{ini} = 5$, $L_{ref} = 25$, $Q = 10000$, $R = 0.001$ - changing them we can tune the behaviour of our controller. Initial trajectory is set to 0.

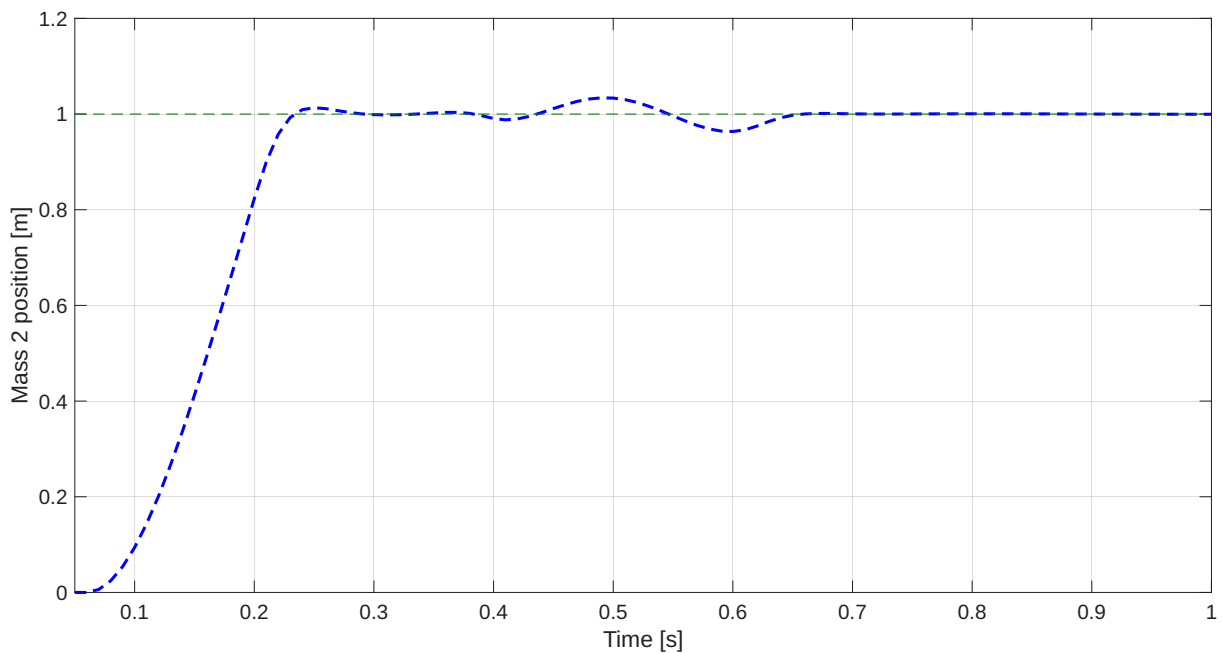


Figure 4.3: Mass 2 position controlled with DeePC

From Figure 4.3, we can see that we were able to reach tracking of the given trajectory; however, it could be tuned for a better result, but as a proof of concept it is sufficient. Next, in Figure 4.4, we display the limited force input.

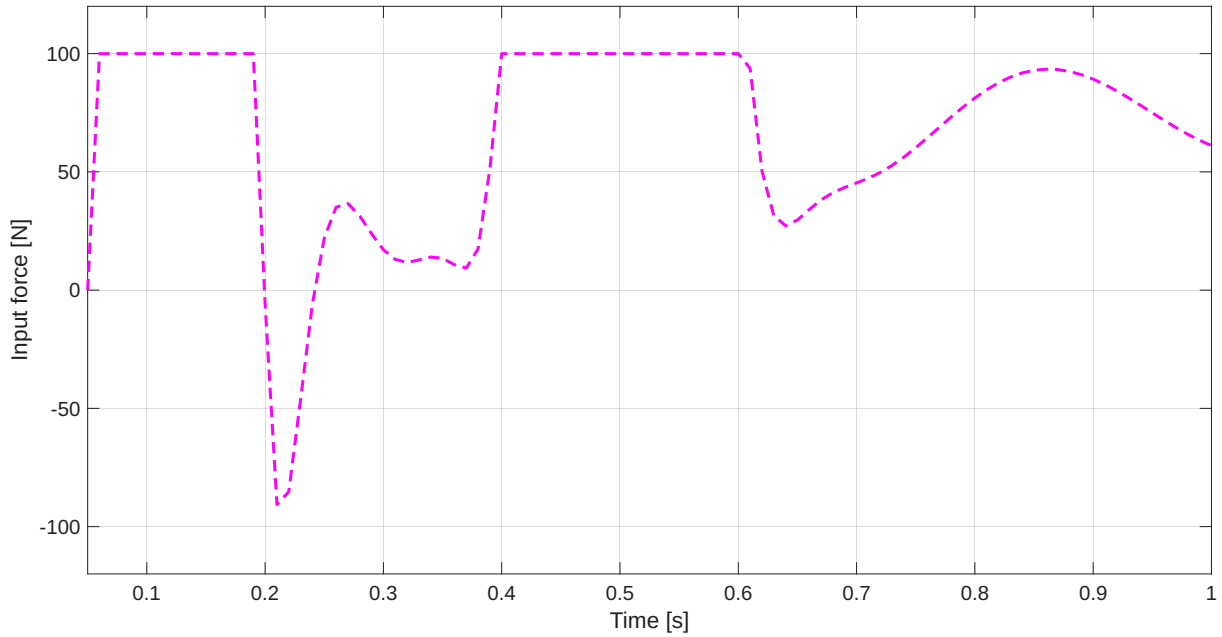


Figure 4.4: Force input for mass 2 position control with DeePC

We can see that the limit on the input is working. Now let's try limiting the position $y \leq 1$ to eliminate overshoot.

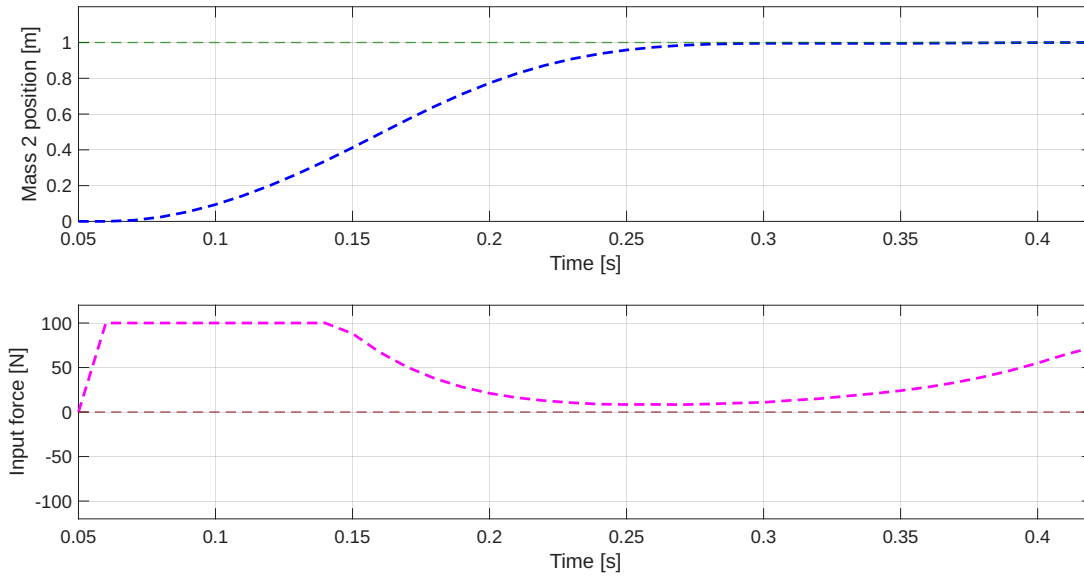


Figure 4.5: Position of mass 2 and force input with DeePC - y limit

Tuning the parameters gave us this result (Figure 4.5), which can be considered relatively good. We got a short settling time with no overshoot. For these simulations, we used MATLAB with YALMIP [16] toolbox and MOSEK [17] solver.

In simulation where systems are completely linear and there is zero noise we get a good result, however, when we consider realistic systems with noise and nonlinearities this DeePC performs poorly - it is not robust. To improve on robustness we implement regularizations to our optimization problem.

4.4 Regularized DeePC

The paper [3] introduced robustifications to the DeePC algorithm (4.2) to make it applicable to realistic systems. We will lay out their results and illustrate them through simulations. The Fundamental lemma is a result which falls apart when we introduce measurement noise (or nonlinearity) to our system since there is a high probability that the image of Hankel matrix of input-output data spans the whole space instead of the subspace which is identified with the behaviour of the system. There is an approach where we can maybe remove the noise from recorded data with SVD or similar techniques. That is good for recorded data but for DeePC where we use real-time data it is not that simple. Now we will formulate regularized DeePC optimization problem:

$$\begin{aligned}
 \min_{g, y, u, \sigma_y} \quad & \sum_{j=0}^{L_{ref}-1} \left(\|\bar{y}_j - r_{t+j}\|_Q^2 + \|\bar{u}_j\|_R^2 \right) \\
 & + \lambda_g \|g\|_2^2 + \lambda_y \|\sigma_y\|_2^2 \\
 \text{subject to:} \quad & \begin{bmatrix} U_p \\ Y_p \\ U_f \\ Y_f \end{bmatrix} g = \begin{bmatrix} \bar{u}_{ini} \\ \bar{y}_{ini} + \sigma_y \\ \bar{u} \\ \bar{y} \end{bmatrix} \\
 & \bar{y}_j \in \mathcal{Y}, \quad j = 0, \dots, L_{ref} - 1; \\
 & \bar{u}_j \in \mathcal{U}, \quad j = 0, \dots, L_{ref} - 1;
 \end{aligned} \tag{4.3}$$

The hard equality constraint (i.e., the equality representing data-based prediction) in the original problem can cause problems when noise is added, so we impose a soft constraint with slack variable $\sigma_y \in \mathbb{R}^{pL_{ini}}$ to ensure the feasibility of the constraint. We penalize the 2-norm squared of the slack variable with the weight $\lambda_y \in \mathbb{R}$, while choosing sufficiently large λ_y "ensures" that $\sigma_y \neq 0$ only if the constraint is infeasible, i.e., when the data is inconsistent. Two-norm squared regularization on g with weight $\lambda_g \in \mathbb{R}$ is motivated by techniques from robust optimization; see, e.g., [ref] for more details. In the original paper [3] 1-norm regularization was used, but in a more recent paper [18] it was updated to 2-norm regularization.

4.4.1 Noisy measurements LTI example

We test the regularized DeePC on 2.1. Since this example requires a non-zero input for a non-zero steady state we replace $\|\bar{u}_j\|_R^2$ with $\|\bar{u}_j - u_s\|_R^2$ in (4.3), where $u_s \in \mathbb{R}^m$ is the steady-state input. This steady-state input is, generally, not necessary for DeePC and it can be calculated from data. We calculate the steady-state input from the model by solving for u_s :

$$\begin{aligned}
 (I - A_d)x_s - B_d u_s &= 0, \\
 Cx_s + Du_s &= y_s,
 \end{aligned} \tag{4.4}$$

where y_s is the desired steady-state output. In our example we have $y_s = 1 \text{ m}$ and solving (4.4) yields $u_s \approx 71.94 \text{ N}$. We introduce Gaussian measurement noise to the system model like this:

$$y_k = Cx_k + Du_k + v_k, \quad v_k \sim \mathcal{N}(0, \sigma^2). \tag{4.5}$$

For this example we took $\sigma = 0.01 \text{ m}$. Now we run regularized DeePC algorithm to track the steady state.

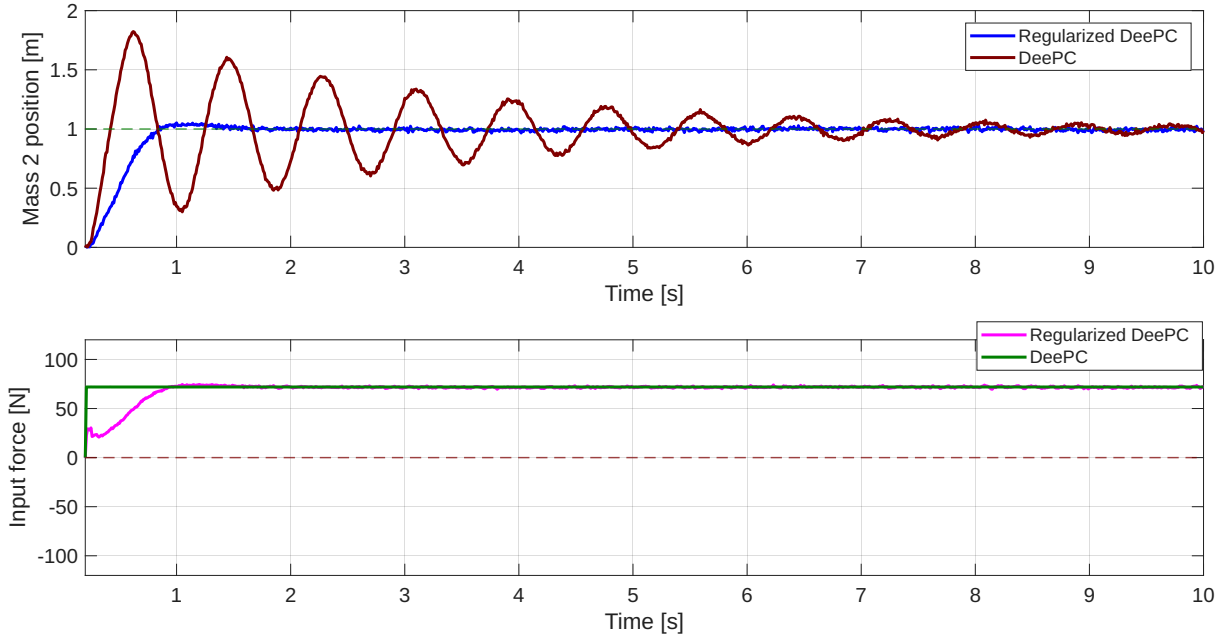


Figure 4.6: Regularized DeePC vs. DeePC; Noisy measurements LTI system

Tuning the parameters of (4.3) gave us a really good result for regularized DeePC and also we can see that DeePC without regularizations performs poorly with the same parameters and data.

4.4.2 Nonlinear system example

Since DeePC regularizations improve performance when noise is added, in this subsection we will test its performance on a mildly nonlinear system with measurement noise. We modify the double spring damper system (2.1) such that the second spring has a nonlinear force-displacement relationship. Concretely, the force exerted by the nonlinear spring is given by:

$$F_2(\delta l) = k_2 \delta l + k_{23}(\delta l)^3. \quad (4.6)$$

Where δl is the displacement, and $k_{23} > 0$ the nonlinear spring constant. We chose $k_{23} = 40 \frac{N}{m^3}$, while all other parameters are unchanged from 2.1. Thus, the nonlinear dynamics can be written as:

$$\begin{aligned} \dot{x} &= Ax + Bu + E(x_2 - x_1)^3 = f(x, u), \\ y &= Cx + Du + v_k, \quad v_k \sim \mathcal{N}(0, \sigma^2). \end{aligned} \quad (4.7)$$

Where $x = (x_1, \dot{x}_1, x_2, \dot{x}_2)^T$ is the state vector, A and B are identical as in (2.16), $C = [0 \ 0 \ 1 \ 0]$,

$D = 0$, $E = [0 \ \frac{k_{23}}{m_1} \ 0 \ -\frac{k_{23}}{m_2}]^T$. Since this is in continuous state-space format, we again discretize it using the Runge-Kutta 4 method (3.4). The nonlinearity is largely expressed for larger displacements, and system is *de facto* linear for small displacement which can be deduced from (4.6), i.e., $(\delta l)^3 \approx 0$ for small enough δl . Choosing k_{23} we can tune how much nonlinearity there will be. It is possible for $k_{23} < 0$; then the spring becomes less stiff for larger displacements. Next, we commence the same procedure for designing a DeePC controller which should track the reference. The steady-state input for $y = 1 \text{ m}$ can be calculated by solving (4.7); we get $u_s \approx 72.96 \text{ N}$.

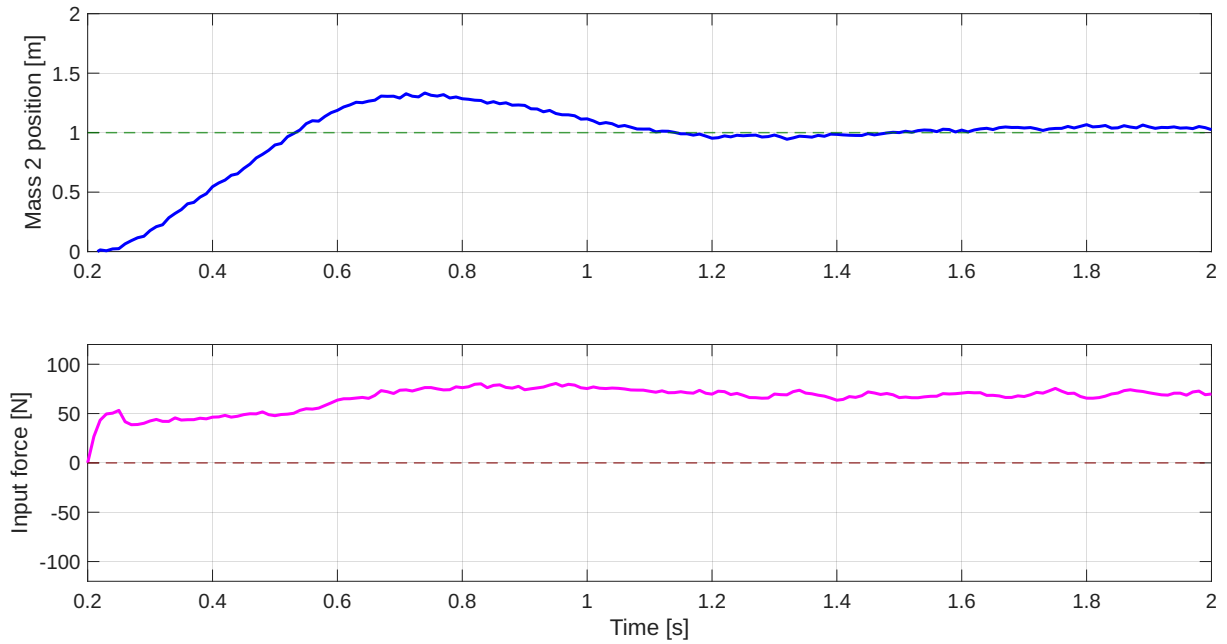


Figure 4.7: Regularized DeePC on a nonlinear system

Figure 4.7 shows that regularized DeePC can track the reference even when the system is nonlinear with noisy measurements. The result is not as good as for LTI systems, but when it's considered that it is completely data-based we can say that the result is good. While doing the simulations we have seen that noise has a more noticeable effect when the underlying system is nonlinear. For improved behaviour, [7] and [19] suggest significant changes to the DeePC algorithm. Also, Koopman operator approach [20] is being researched for potential improvements for (data-based) nonlinear system control.

5 Conclusion

In this thesis, Willems' Fundamental Lemma was presented from both a theoretical and an applied perspective. The necessary theoretical background was introduced and a proof of the Lemma was provided. Subsequently, the lemma was validated on several numerical examples and used for data-based simulation of the future trajectories of linear systems. The applicability of the Lemma for simulating the behaviour of nonlinear systems was investigated. Furthermore, a data-based predictive control algorithm was formulated, which incorporates the Lemma within a suitable optimization problem to control linear systems. Regularization terms were introduced to make the controller more robust to working with realistic systems affected by noise and nonlinearities. Future work could focus on further examining the theoretical aspects of the Lemma and exploring a broader range of applications, as well as on experimental validation of the data-based control methods on a real-world system. Overall, the wide range of applications enabled by the Willems' Fundamental Lemma justifies its fundamental role in modern data-based systems and control theory.

A Appendix

Link to the GitHub repository containing all MATLAB code used in this thesis:

<https://github.com/AndrijaAdamovic/bachelors-thesis>

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